Assignment 3 - MTH3230

Next-Day Maximum Temperature Forecasting at Melbourne Airport Using a GRU

Problem statement

To predict the following day's maximum temperature using the previous 30 days' maximum temperatures.

Potential users: people planning outdoor activities or temperature-sensitive work near or at Melbourne Airport.
Success criterion: to outperform the naive baseline on MAE and RMSE.

Dataset Description

Dataset: https://www.kaggle.com/datasets/jsphyg/weather-dataset-rattle-package

This dataset contains about 10 years of daily weather observations from numerous Australian weather stations.

Target Variable = 'MaxTemp' column in the dataset for the 'MelbourneAirport' station.
Sampling frequency = 'MaxTemp' recorded daily

The entire dataset consisted of many stations within Australia.
Since the task asked for the prediction of a continuous variable for one station, I picked the 'MelbourneAirport' station.

The next step involved reducing the dataset to my required specifications which were:
'Location' = 'MelbourneAirport'
'MaxTemp' columns only.

Item	Description
Observed temperatures	3,009
Calendar days after reindexing	3,098
Missing days	89
Training period	1 January 2009–8 December 2014
Validation period	9 December 2014–17 March 2016
Test period	18 March 2016–25 June 2017

The above is a summary of the dataset with the MelbourneAirport' station selected only.

While preprocessing the data , I noticed that a few dates were missing between the first date of observation and the final date of observation.
The next step to be taken was to include these dates as indexes to the dataset and to give them NaN values since we did not know the MaxTemp values for these dates.

If these indexes were missing in our dataset it would have led to poor spacing for the values in our dataset.

The next step involved dividing the dataset into training, validation and testing subsets.
The split was chronological, approximately 70% training, 15% validation and 15% test.
The training subset was used to train the GRU model.
The validation subset was used to test the GRU model across various window sizes. It was also used to validate naive predictions as well.
The testing subset consisted of unseen data , the model never saw to test final behavior and accuracy of the model.

After this , I used a standard scaler to scale the target variable ['MaxTemp'] to its standardized scores. These values resemble its 'z' score.
This was done because Neural networks prefer input variables that are centered on 0 for its training and optimization.
The scaler was fitted only on observed training temperatures, then applied unchanged to validation and test data.
Predictions were inverse-transformed into °C before evaluation.

The next step involved defining a function to create sliding windows from the scaled temperature series. For a specified window length, each input consists of that number of consecutive daily observations, while the target is the following day's scaled maximum temperature. The window advances by one day to generate each subsequent input–target pair. Any pair containing a missing input or target value is excluded.

Model design summary

Since the task involved a continuous ordered sequence of values, I chose a GRU as an initial starting point.
GRU's from what I researched are able to gain and store information from previous history in its memory as a hidden state.
A GRU can learn how much of the previous information to retain and how much of the new information has to be incorporated into it.

That makes it suitable for learning patterns such as temperature persistence, recent warming or cooling, and relationships across several days.
Compared with an LSTM with the same input and hidden dimensions, a GRU has fewer parameters. This makes it more suitable as a starting point for training a smaller dataset.

Architecture: One GRU layer with 32 hidden units and a linear output. Chosen to learn patterns in sequential temperatures while keeping the model small.

Hyperparameters: Tested 7-, 14- and 30-day windows; selected 30 days for its lowest validation RMSE. Batch size 32, learning rate 0.001 and maximum 100 epochs were practical starting choices.

Optimizer: Adam. Chosen for its adaptive weight updates and suitability as a starting optimizer.

Activation functions: Standard GRU sigmoid gates control information retention, while tanh forms candidate memory values. A linear output allows unrestricted continuous predictions.

Regularization: Early stopping with patience of 10 epochs limits overfitting and restores the best validation weights. No dropout or weight decay was used.

Naive Prediction: A naive persistence forecast was used as a baseline, predicting the next day's maximum temperature as equal to the current day's observed maximum temperature. This method requires no training and provides a simple benchmark for assessing whether the GRU improves forecasting accuracy. Both methods were evaluated on the same target dates using MAE and RMSE in degrees Celsius.

Performance Results

Validation Comparison table:

```
...
    Training GRU with a 7-day input window
    Epoch   1 | Train MSE: 0.6481 | Validation MSE: 0.5980
    Epoch  10 | Train MSE: 0.3621 | Validation MSE: 0.5245
    Epoch  20 | Train MSE: 0.3578 | Validation MSE: 0.5265
    Epoch  30 | Train MSE: 0.3560 | Validation MSE: 0.5236
    Early stopping at epoch 34

    Training GRU with a 14-day input window
    Epoch   1 | Train MSE: 0.6262 | Validation MSE: 0.5708
    Epoch  10 | Train MSE: 0.3507 | Validation MSE: 0.5158
    Epoch  20 | Train MSE: 0.3463 | Validation MSE: 0.5047
    Epoch  30 | Train MSE: 0.3444 | Validation MSE: 0.5270
    Early stopping at epoch 30

    Training GRU with a 30-day input window
    Epoch   1 | Train MSE: 0.6176 | Validation MSE: 0.5694
    Epoch  10 | Train MSE: 0.3394 | Validation MSE: 0.5079
    Epoch  20 | Train MSE: 0.3308 | Validation MSE: 0.5003
    Epoch  30 | Train MSE: 0.3264 | Validation MSE: 0.5096
    Early stopping at epoch 30

    Validation comparison — errors in °C
```

Window	Best epoch	GRU MAE	GRU RMSE	Naive MAE	Naive RMSE
30	20	3.508	4.608	4.044	5.548
14	20	3.516	4.628	4.044	5.548
7	24	3.581	4.700	4.044	5.548

```
...
    Selected input window: 30 days
```

The above image consists of the aggregated results of the GRU trained on various window sizes against the validation set.
We can also compare it to the naive prediction errors.

As seen by the results in the table, A window of size = 30 showed the least error. Hence that was chosen.

Final Performance table:

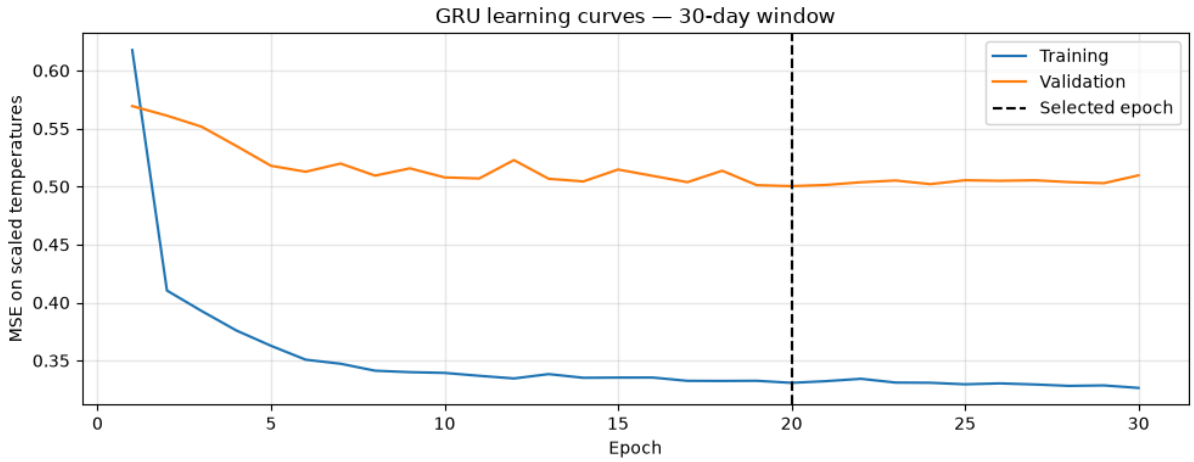
Errors in °C

	GRU MAE	GRU RMSE	Naive MAE	Naive RMSE
train	2.802	3.752	3.086	4.366
validation	3.508	4.608	4.044	5.548
test	2.768	3.673	3.134	4.359

Test RMSE improvement over naive: 15.74%

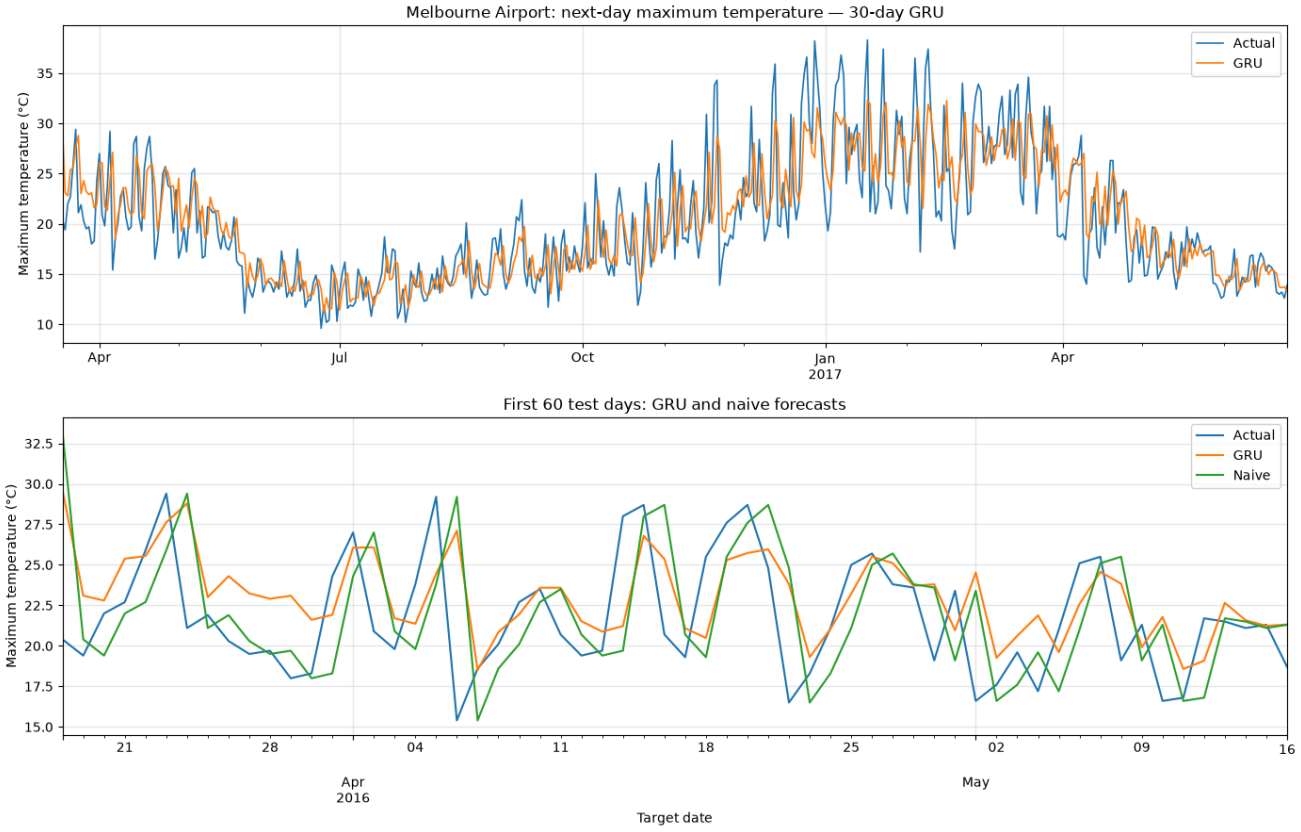
As seen in the above table, the GRU achieved test MAE = 2.768°C and test RMSE = 3.673°C, reducing RMSE by 15.74% compared with the naive forecast.

Learning-curve plot:



We can see that validation loss stopped improving substantially, with the best weights at epoch 20. Early stopping ended training at epoch 30 and restored those weights.

Test forecast plots:



The model follows the broad temperature pattern but misses some sharp changes and extreme peaks.

Interpretation, limitations, and possible next steps

Interpretation:
The selected 30-day GRU outperformed the naive baseline, achieving a test MAE of 2.768°C and RMSE of 3.673°C. This represents a 15.74% reduction in RMSE compared with the baseline.
The forecast plots show that the model follows the broad seasonal temperature pattern, although its predictions are smoother than the observations and miss some sudden changes and extreme peaks.

Limitations:

1. The model uses only past maximum temperatures, excluding other weather information that may influence future temperatures.

2. Missing observations reduced the number of usable training samples.
3. The experiments compared three input-window lengths while keeping other hyperparameters fixed.
4. Results are limited to one station and one historical test period, so they do not establish performance at other locations or under current weather conditions.

Possible Next Steps:

1. Future work could investigate different hidden-layer sizes, learning rates and additional forecasting architectures.
2. Evaluation across multiple time periods and random seeds would help assess the consistency of the results.

Overall, the GRU improved next-day maximum temperature forecasts relative to the naive baseline in this experiment, with the 30-day input window achieving the lowest validation RMSE among the configurations tested.